

CausalCache: Conditional High-Fidelity Restoration for Long-Horizon GUI Agents

Supplementary Material

Anonymous submission

Scope of the Supplement

Arm naming in this supplement is explicit: HGKV+selector denotes the adapter-attached selector arm (three Mobile-World rounds); the deployed CAUSALCACHE of the main text is the frozen policy + selector (single round). Recent/summary arms are shared.

The main paper is self-contained. This supplementary document provides the complete fixed-budget design, confidence intervals for the policy and selector ablations, per-round closed-loop results, and deployment-cost measurements. The supplement is a separate PDF and does not count toward the seven pages of main content. All results use the same summary trace and archived screenshots as the main paper. Restoring an event means retrieving its archived post-action screenshot and reattaching it to the event summary; no screenshot is generated from text.

Fixed-Budget Restoration Protocol

At decision t , every completed event remains visible in text. An active visual-context budget B allows exactly B events to occupy the official retained-turn form: the step’s verbatim policy response paired with its reattached archived screenshot. Every other event appears as a one-line action summary. Recent- B assigns the retained turns to the latest B eligible events. A restoration allocation may instead demote a recent event to its one-line summary and promote a distant event to its retained turn, preserving the image count, image resolution, retained-turn count, and prompt structure; the official protocol pairs each retained observation with the response that produced it, so the two forms move as a unit. The realized number of non-recent promotions is

$$k = |S \setminus \text{Recent-}B|.$$

It is an observed selector output, not a preset experimental treatment.

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B	Recent arm	Replacement training arm	Deployed allocation
1	Recent-1	recent $\rightarrow$ archived	exact-1
2	Recent-2	oldest $\rightarrow$ archived	exact-2 (recent fallback)
4	Recent-4	oldest $\rightarrow$ archived	exact-4 (recent fallback)
8	Recent-8	built, not trained	exact-8 (recent fallback)

Table S1: Matched-budget fidelity-restoration design. The $B=8$ row is a budget-extrapolation test; it is excluded from adapter and selector training.

Intervention arms. For a matched group, R_0 and R_A use Recent- B under the frozen and adapted policies; S_0 and S_A replace the oldest recent image with a target-relevant archived image; W_A uses an age-matched wrong archived image. The adapter selection statistic is

$$\text{DiD}_{\text{select}} = (S_A - S_0) - (R_A - R_0).$$

The gate requires a positive episode-clustered bootstrap lower bound and point-estimate drift caps $|R_A - R_0| < 0.02$ and $|W_A - W_0| < 0.02$. Reported coalition utilities always rerun the complete restored set; they are never sums of singleton utilities.

Single-slot training interface ($k=1$). Under the strictest *image-only* criterion—the required value appears in neither the goal, the current frame, the recent window, nor any completed step’s action text—mining 1,326 GUI-Odyssey trajectories yields 96 image-only decision points at $B=4$, of which only two require two distinct evidence frames simultaneously (3 of 239 at $B=1$); on the desktop corpus, 75% of eligible states admit exactly one recurrence witness. A matched-budget probe under the official multi-turn protocol shows the first replacement is worth its slot ($+0.013$, 95% CI $[+0.005, +0.021]$) while the paired increments of a second and third replacement are indistinguishable from zero or negative ($+0.001$ and -0.011 , n.s.). Supervision is annotation-free, so a domain exhibiting multi-frame demand requires only regenerating the corpus with $k \geq 2$ restoration arms: a data-mix change, not a method change.

Uncertainty. Offline intervals use 2,000 episode-clustered bootstrap replicates. Closed-loop intervals resample tasks while preserving the paired arms and repeated-run structure. A confidence interval crossing zero is described as indistinguishable, not as evidence for equivalence.

Additional Desktop Results

Policy-Side Gates Across Visual Budgets

Adapter	B	DiD _{select} [95% CI]	$ A_r $	Wrong	Gate
HGKV	1	+0.0224 [+0.0162, +0.0294]	.0088	.0085	pass
HGKV	2	+0.0164 [+0.0110, +0.0221]	.0107	.0078	pass
HGKV	4	+0.0109 [+0.0070, +0.0154]	.0098	.0091	pass
HGKV	8	+0.0063 [+0.0018, +0.0115]	.0106	.0137	pass
Ungated	1	+0.0177 [+0.0135, +0.0220]	.0086	.0071	pass
Ungated	2	+0.0139 [+0.0102, +0.0177]	.0079	.0069	pass
Ungated	4	+0.0075 [+0.0031, +0.0118]	.0098	.0089	pass
Ungated	8	+0.0047 [+0.0007, +0.0093]	.0160	.0163	pass

Table S3: Complete per-budget adapter gates. Group counts for $B = 1/2/4/8$ are 94/72/45/11; the frozen selection effects are +0.0791/ + .0256/ - .0002/ - .0305. $B=8$ is untrained.

Both adapters pass, but HGKV has a larger selection effect at every budget and remains farther from the drift caps at $B=8$. Across the trained budgets $B \in \{1, 2, 4\}$, HGKV maintains positive selectivity and the learned allocation improves over Recent- B at the deployment-relevant budgets; at the unseen $B=8$, policy-side selectivity transfers, whereas the end-to-end offline allocation gain is inconclusive.

Why the recent reference matters. (The probes in this paragraph were measured during the initial development iteration of the corpus and adapter; the frozen-policy contrasts are checkpoint-independent, and the adapter re-scoring value is from that iteration’s selected checkpoint.) At $B=1$, the deployment renderer’s most recent slot can duplicate the current screenshot. Against that redundant control, the frozen policy prefers the relevant archived event by +0.0333 nats/token (95% CI [+0.0118, +0.0567]). Against the informative true previous frame, the frozen preference is only +0.0013 [-0.0242, +0.0253]. The HGKV adapter retains a positive true-previous-frame DiD of +0.0095 [+0.0056, +0.0138]. Thus the learned selectivity is not an artifact of comparing useful pixels with a duplicated current screen.

Selector and Feature Ablations

Allocation rule	$B=2$	$B=4$
	+0.0297	+0.0169
Learned scorer	[+0.0185, +0.0409] +0.0058	[+0.0067, +0.0277] +0.0082
Random exact- B	[-0.0056, +0.0168] +0.0061	[-0.0009, +0.0177] +0.0071
RGB similarity	[-0.0032, +0.0150] -0.0159	[-0.0010, +0.0152] -0.0091
Inverted score	[-0.0290, -0.0033] +0.0121	[-0.0180, -0.0004] +0.0084
No recency features	[-0.0003, +0.0246]	[-0.0012, +0.0173]

Table S4: Exact- B selector ablations with fresh full-set policy scoring. Random and RGB-similarity rules do not explain the gain; reversing the learned ranking is harmful.

At $B=4$, the learned scorer chooses $k = 0/1/2/3/4$ on 103/103/88/119/146 evaluated states. It therefore retains the entire recent window on 19% of states and performs

non-recent restoration selectively. The corresponding distributions for random allocation are 96/138/161/152/48, and for RGB similarity 233/166/106/66/24. Performance cannot be attributed to using a fixed sparse history pattern or to always displacing recent images.

Selector input features. The scorer’s input encodes, explicitly: candidate age and recency-window membership, target-recurrence witness signals with coordinate distance, action family, frame-deduplication multiplicity, trajectory shape, and set context (selected-set size, overlap and age gaps between the candidate and the already-selected events). The witness features compare the action taken right after an archived event against the reference action.

Single-pass reference variant. Replacing the proposal reference with the previously executed action removes the second policy call, adding only the selector’s own cost (< 10 ms). Offline it matches the proposal reference (+0.039/ + .025/ + .015/ - .005 over Recent- B at $B=1/2/4/8$) and on OSWorld-Verified it reaches 34.1%; zero-shot on mobile it drops to 29.1% against 33.9% for the deployed proposal-conditioned method, with the failure concentrated on single-app control tasks where the last-action reference trivially recurs. Cross-platform transfer is what the proposal reference buys.

Witness-reference ladder. The single-pass last-action reference yields +0.0247 [+0.0133, +0.0363] at $B=2$ and +0.0152 [+0.0058, +0.0250] at $B=4$ on desktop. Removing the witness reduces these effects to +0.0161 [+0.0025, +0.0299] and +0.0086 [-0.0024, +0.0196], respectively. This motivates the proposal-conditioned two-pass path as the default for cross-platform transfer, while retaining last-action conditioning as a domain-matched efficiency variant.

Additional Closed-Loop Results

Zero-Shot Mobile Evaluation

Memory-critical split construction. The split is fixed by a mechanical rule over the benchmark’s own task definitions: `memory_candidate = tasks whose official metadata lists more than one application (app_count  $\geq 2$  from the upstream BaseTask.app_names)`, `single_app_control = exactly one`, restricted to the GUI-only interface. Every record in the frozen manifest carries the upstream source path and content hash; no model outcome and no manual labeling enter the rule. Cross-application membership is a preregistered memory-challenge *candidate*, not a per-task guarantee of memory dependence: candidates that do not in fact require distant evidence dilute the stratum toward zero, so the reported contrast is conservative.

Method	Run 1	Run 2	Run 3	Mean
Frozen, summary-only	33/117	32/117	34/117	28.2%
HGKV + Recent-4	34/117	37/117	35/117	30.2%
HGKV+selector	39/117	42/117	38/117	33.9%

Fixed-budget fidelity allocation (offline desktop heldout; primary setting $B = 4$)

Policy	Allocation	$B=1$	$B=2$	$B=4$	$B=8^\dagger$	Realized k	Notes
Frozen	Recent- B	0	0	0	0	0	reference
Frozen	oldest slot $\rightarrow$ archived	+0.079	+0.026	−0.000	−0.031	1	
HGKV	Recent- B	+0.002	+0.005	+0.007	+0.009	0	drift-capped
HGKV	oldest slot $\rightarrow$ archived	+0.103	+0.047	+0.018	−0.016	1	gate: main Tab. 3
HGKV	selector, last-action reference	+0.039	+0.025	+0.015	−0.005	measured	71% displace ≥ 1 at $B=4$
HGKV	selector, oracle-reference variant	+0.044	+0.025	+0.011	−0.002	measured	teacher-forced oracle

Table S2: Fixed-budget fidelity-allocation results. Each cell is a matched-count contrast on the desktop heldout split; every allocation promotes exactly B events to summary-plus-image form, and the learned selector may keep every promotion on the recent window ($k=0$); $^\dagger B=8$ is excluded from training and reports budget extrapolation. Cross- B comparisons are analysis only.

Table S5: Per-round zero-shot MobileWorld success. The paired HGKV+selector minus HGKV+Recent-4 mean is +3.7pp (95% CI [+0.8, +10.3]).

Stratum	HGKV+selector	Recent-4	Δ
			+8.6pp
Memory-critical	28.0%	19.4%	[+1.6, +16.1] −1.8pp
Control	40.6%	42.4%	[−6.7, +3.6]

Table S6: Primary mobile allocation analysis ($n = 62$ memory-critical; $n = 55$ control). The split-by-method interaction is +10.4pp (95% CI [+1.5, +19.6], two-sided $p = 0.022$).

Outcome-defined sensitivity analysis. On the 95 tasks where the frozen summary-only policy reaches the official step cap, HGKV+selector scores 32.6%, Recent-4 28.9%, and summary-only 23.9%. The paired HGKV+selector minus summary-only difference is +8.8pp [+1.8, +16.0]; the paired difference from Recent-4 is +3.7pp [−2.6, +10.0]. On the 22 tasks where summary-only terminates early, HGKV+selector is 43.2%, Recent-4 is 36.4%, and summary-only is 47.0%; HGKV+selector minus summary-only is −3.8pp [−20.5, +12.1]. These outcome-defined strata are secondary because they condition on baseline behavior; the construction-defined split in Table S6 is the primary analysis.

Two-pass mobile cost. Across 2,224 audited steps, the proposal-conditioned second policy pass fires on 81.7% of all steps and 96.9% of steps with more than B eligible events. Median pass-1, selector, and conditional pass-2 times are 4.229 s, 0.039 s, and 3.258 s, respectively. Mean per-step overhead relative to pass 1 is 65.9%. The selector itself is cheap; the extra policy pass dominates.

OSWorld-Verified Evaluation

Method	Rate	Paired comparison
B0	19.9% (72)	reference +12.7pp [+8.6, +17.2]
FR	32.7% (118)	vs. B0 +0.3pp [−3.1, +3.6]
HR	33.0% (119)	vs. FR −0.3pp [−3.9, +3.3]
HGKV+selector	32.7% (118)	vs. HR

Table S7: OSWorld-Verified closed-loop results on the official no-GDrive 361-task roster with a 15-step budget; parenthesized values are success counts. High-fidelity history is beneficial relative to summary-only memory; same-budget allocations are statistically indistinguishable. B0 denotes frozen summary-only, FR frozen Recent-4, and HR HGKV Recent-4.

For HGKV+selector on OSWorld, the median task incurs eight extra policy calls and 20.6 model-seconds. Median wall time is 307.6 s, so the added model time is approximately 7% of end-to-end task time. These results separate two claims: desktop closed loop supports the availability of high-fidelity history, while the mobile construction-defined split tests which summarized events should receive that fidelity.

Closed-Loop Budget Sweep

Single-round MobileWorld campaigns evaluate HGKV+selector at $B \in \{1, 2, 8\}$ with the same frozen weights; only the selection budget and the runner’s matched image budget change. The same-budget references are the frozen Recent- B dose curve at $B \in \{1, 2\}$ (licensed by the adapter’s behavioral neutrality on matched allocations, verified on both platforms) and a dedicated HGKV+Recent-8 campaign at $B=8$, which also extends the extrapolation test from the offline gates to the closed loop. $B=4$ remains the primary setting; single-round budget cells are trend evidence, not headline contrasts.

Table S11: 32B DiD gates on the 94 development groups

Arm	Full (117)	Mem.-crit. (62)	Control (35)
Frozen + selector (CAUSALCACHE)	36.8	30.6	43.6
HGKV+selector (3-run mean)	33.9	28.0	40.6
frozen-taught selector	33.3	25.8	41.8
multi-slot-corpus-taught selector	33.3	24.2	43.8
$k \leq 1$ -capped selector	28.2	19.4	38.2
OCR-text restoration	32.5	24.2	41.8
verbatim-text restoration	27.6	18.0	38.2
Recent-4 (3-run mean)	30.2	19.4	42.4

Table S10: All deployment arms at the primary budget. Task-paired bootstrap contrasts (20,000 replicates): CAUSALCACHE vs. Recent-4 full +6.6pp [+0.3, +13.1] ($p=0.035$), memory-critical +11.3pp [+2.7, +20.4] ($p=0.006$); CAUSALCACHE vs. HGKV+selector full +2.9pp [−2.9, +8.6] (indistinguishable); $k \leq 1$ cap vs. HGKV+selector memory-critical −8.6pp [−16.7, −1.6] ($p=0.02$); verbatim text vs. HGKV+selector memory-critical −10.4pp [−18.6, −3.3] ($p=0.006$); OCR vs. Recent-4 memory-critical +4.8pp [−2.2, +12.4].

Teacher Ablations

The deployed selector distills set utilities scored under the HGKV instrument (all 5,335 states; 58k singleton + 147k set edges). Two retrained variants swap only the label teacher, with set labels rescored on a uniformly interleaved 60% state subset (identical tree parameters): scoring under the fully frozen policy yields 25.8% memory-critical—still significantly above Recent-4 (+6.5pp, $p=0.04$)—and scoring under an HGKV variant retrained on selector-composed multi-slot corpora yields 24.2%. The HGKV-taught deployed selector reaches 30.6%; because the retrained variants also use the reduced label pool, the gap conflates teacher model with label volume, and the matched-volume comparison (frozen vs. multi-slot-taught, both 60%) shows no teacher separation. The instrument’s necessity is therefore established offline (Table 2 of the main text: the frozen policy carries no measurable selectivity), while its closed-loop teaching advantage over a frozen scorer at matched label volume remains open.

Cross-Backbone Gates (GUI-Owl-1.5-32B)

The v4 recipe—same corpus, objective, 300 steps, checkpoint cadence 50—applied unchanged to the 32B backbone (world size 3, effective batch 6):

Checkpoint	DiD select	95% CI	A_r
step 50	+0.0009	[−0.0002, +0.0020]	+0.0005
step 100	+0.0067	[+0.0048, +0.0088]	−0.0023
step 150	+0.0049	[+0.0031, +0.0068]	+0.0014
step 200 (selected)	+0.0142	[+0.0109, +0.0176]	−0.0063
step 250	+0.0126	[+0.0099, +0.0155]	−0.0029
step 300	+0.0126	[+0.0097, +0.0156]	−0.0017

($B=1$). Five of six checkpoints pass all pre-specified gates; the frozen selection rule picks step 200. Recent-window drift stays within a third of the 0.02 cap throughout: token-gated selectivity inside the drift envelope transfers across backbone scale without retuning.